

Network Analysis & Circuits

BRANCH: ELECTRICAL (EE)

COURSE CODE: EE201

Subject Overview & Scope

This comprehensive document serves as the official compiled study notes for Network Analysis & Circuits (EE201) under the Electrical (EE) department. It is structured into four core university syllabus units covering primary concepts, definitions, formulas, and advanced technical applications for university examinations and placements.

University Course Syllabus & Core Units

Unit 1: Network Theorems

KCL, KVL. Nodal and Mesh analysis. Superposition, Thevenin's, Norton's, Maximum Power Transfer, Reciprocity, and Millman's Theorems.

Unit 2: Transient Analysis

Steady state and transient response. Analysis of RL, RC, RLC circuits under DC excitation. Initial conditions, Laplace transform application.

Unit 3: Resonance & Two-Port Networks

Series and Parallel resonance: bandwidth, Q-factor, selectivity. Two-port networks parameters: Z, Y, ABCD, h parameters, conversions.

Unit 4: AC Circuits & Filters

Phasors, impedance, power triangle, power factor. Three-phase balanced/unbalanced circuits. Passive filters: Low pass, High pass, Band pass, Band stop.

*Thevenin's voltage V_{th} , Norton's current $I_n = V_{th} / R_{th}$. Series resonance frequency: $f_r = 1 / (2 * \pi * \sqrt{L * C})$.*

CRITICAL FORMULA / PRINCIPLE: Key Formulas & Exam Pointers



